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Retrieval-Augmented Simulacra:
Generative Agents for Up-to-date and
Knowledge-Adaptive Simulations

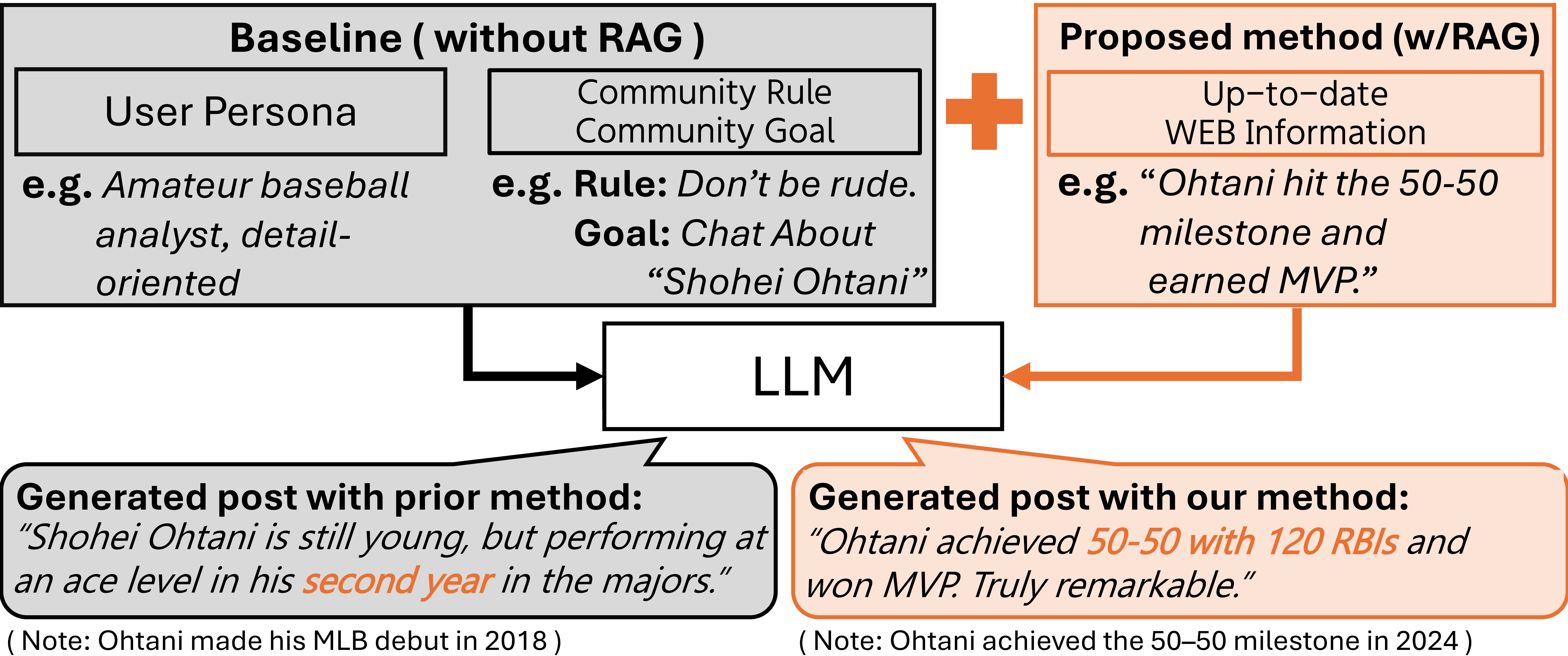
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What for?

Real social media discussions are complex and hard to predict.
We simulate conversations with LLMs to anticipate how discussions may unfold.

How?

LLMs generate posts from a user persona(short description of traits),
a community goal(topic), community rules, and prior posts.
We also provide web information so that generated posts reflect up-to-date
content, enabling realistic SNS simulations.



Result / Discussion

Evaluation method

Crowdsourcing survey on generated
posts / threads.

Results

Our method improved naturalness for
Rinna Youko 70B and GPT-4o.
For Gemini, both “most natural” and
“most unnatural” decreased. The
proposed method tends to stabilize
evaluation distribution.

Discussion

Effectiveness depends on model
capability:
- Works well with high-capability
models like GPT-4o.
- May suppress peak naturalness for
mid-level models like Gemini.
Balancing stability and peak
naturalness remains a challenge.

